Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/21/2017
PI Name:	Oates, Chris Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2018 per M. Lee/NASA HQ (Ed., 2/16/16)		
Key Personnel Changes/Previous PI:	EDITOR'S NOTE (December 2015): PI Chris Oates took over the project from Thomas O'Brian on 11/16/2015.		
COI Name (Institution):	Jefferts, Steven Ph.D. (National Institute of Standards & Technology)		
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EDITOR'S NOTE (December 2015): Principal Investigator (PI) Chris Oates took over the project from Thomas O'Brian on 11/16/2015.
To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to:

1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.

Task Description:	2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources. As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following: • Global time and frequency transfer with uncertainty as low as 10-17. • Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10-17 or better. • Test of gravitation redshift at 2 x 10-6 uncertainty or better. • Search for variation in fundamental constants, such as fine structure constant at level of 10-18 per year or better. All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list): • World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10-16 or better. • Two mercury ion optical clocks with current fractional frequency uncertainty of approximately 1 x 10-17 and continually improving. • Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 8 x 10-18 and continually improving. • Two ytterbium lattice optical clocks with uncertainties better than 1 x 10-16 and rapidly improving. • Two strontium lattice optical clocks with uncertainties approximately 3 x 10-17 and rapidly improving. (At JILA. A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.) • A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons. • A world-class time scale comprising seven hydrogen masers, with eight more masers to be added over the next three years. NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications. In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment. We propose the following revised program plan outline: Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction. Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure. Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time. Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time. Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time. Gap Year (2017): Perform relativistic calculations to prevent unanticipated shifts from contaminating ACES data. Install updated MWL hardware into new NIST time and frequency transfer system. Prepare updated data analysis system in anticipation of ACES launch. $150k for partial support of staff time. Gap 6 Mo. (2018): Assuming successful launch, participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $80k for partial support of staff time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major benefits and impacts of this research program can be grouped into three main areas: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from

ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory.

3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.

Task Progress:

In the most recent reporting period (November 2016 through November 2017) work has focused on two main tasks:
1. Preparation of the NIST site for the Microwave Link Ground Terminal (MWL GT) including progress on the NIST microwave and optical clocks.

2. Advancing the NIST time and frequency infrastructure time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics.

3. Designing simulated data for testing the NIST data processing software in preparation for analyzing real data transmitted from the clocks onboard the ISS including relativistic effects and time delays.

Progress on each task is described below.

1. Preparation of the NIST site for the Microwave Link Ground Terminal (MWL GT) including progress on the NIST microwave and optical clocks

The basis for the ACES program will be the advanced primary frequency standard on the International Space Station (ISS), which will link to participating microwave ground stations around the world. NIST anticipates receiving its Microwave Link Ground Terminal (MWL GT) from ESA in early 2018 in anticipation of a 2018 ACES launch. NIST has been closely coordinating with ESA on details of the MWL GT hardware, information technology and infrastructure requirements.

During the reporting period ESA officials visited NIST to discuss shipping/hardware details and inspect the NIST Ground Terminal platform, which NIST installed on the roof of its Precision Measurement Laboratory in Boulder. NIST clock scientists also worked with NIST services to prepare for installation of the MWL GT including identifying the logistics for getting the GT on the roof and obtaining the necessary transmission license.

Another key aspect in preparation for the 2018 launch is the installation of the requisite information technology (IT) infrastructure. We have been working closely with CADMOS (Center for Development of Microgravity and Space Operations), who is responsible for the ESA IT on this project. Data files will be exchanged between NIST and ACES headquarters through a data connection between the Boulder Ground Terminal and User Support and Operations (USOC), located in Toulouse, France. In preparation for this exchange, we have adopted the format of the clock status files that needs to be uploaded to the ACES computer connected to the MWL and have written scripts to automate the data upload process.

2. Advancing the NIST time and frequency infrastructure time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics

NIST continues to improve the performance of its multiple optical and microwave frequency standards, which will be crucial to international time and frequency comparisons and ACES-mediated fundamental research. The past project year included advances in several of NIST's frequency standards. NIST's Yb lattice clocks and Al+ logic clocks both had key systematic uncertainties reduced to the point that they have both demonstrated an total fractional frequency uncertainty below 2 x 10-18, competitive with the best clocks in the world. Additionally, the Yb clock has achieved a stability that enables it to reach the 1 x 10-18 level in a few hours of averaging time, which enables these clocks to be exceptionally precise standards for fundamental physics research. This high level of stability dramatically advances the goals of ACES time and frequency transfer given the relatively short visibility of the ISS to any MWL GT.

The NIST Calcium Thermal Beam Clock demonstrated a fractional frequency instability of 4 x 10-16 at one second averaging time, more than a factor of 20 lower than any other thermal beam standard worldwide. Finally, NIST has initiated measurements between the NIST Time Scale and its optical standards with the goal of eventually incorporating the optical standards into the Time Scale. This would lead to a more accurate and stable Time Scale.

At the time of this report, NIST (including NIST/JILA) currently operates two of the best microwave primary frequency standards (Cs fountains) and five leading optical standards (Yb and Sr neutral atom lattice clocks, Hg+ and Al ion clocks, and a Ca thermal beam clock). Other laboratories across the world are achieving high performance as well, and the NIST optical frequency standards are likely to continue to improve. This dramatic improvement of optical frequency standard performance strongly supports ACES goals of improved international time and frequency transfer and the ability to improve fundamental tests of physics.

3. Designing and testing the NIST data processing software in preparation for analyzing real data transmitted from the clocks onboard the ISS

After installation of the Ground Station itself, perhaps the most important task in preparation for the ACES launch is to make sure that we are in agreement with the ACES team on all the necessary corrections for the frequency transfer from the ISS to our Ground Terminal. To this end, we have worked with mock data provided by an ESA portal, which in the past has revealed some discrepancies between our results and those of CADMOS, who is overseeing the ACES data transfer. During this reporting period we have addressed two key issues regarding potential corrections: (1) the possible desynchronization between the ISS and ground clocks, and (2) the physics behind gravitational/relativistic effects. Part of this effort has included the production of data sets in the ACES format (using geocentric coordinate time), and have added ionosphere, tropospheric, and instrument delays to simulate the data sets as accurately as possible.

Our models have found the desynchronization changes with time because the proper time on the orbiting clock is

	dominated by time dilation; the speed of the orbiting clock is nearly escape speed while the altitude is not very great so there is not much gravitational blue shift. Thus, we have added a non-zero desynchronization to the models. We have continued to investigate gravitational/relativistic effects and have now recast the uplink-downlink sequence of signal transmissions in a form with solutions yielding closed form expressions involving third order terms in 1/c, which we believe is a more accurate way of describing frequency transformation. Another issue of concern regards the displacement of the surface of the Earth due to Earth tides. This effect needs to be studied in detail at the location of NIST in order to compute the true range between NIST and the orbiting ISS. Using GPS carrier phase data, we have found that this change in the earth surface can be precisely determined within a few millimeter accuracy. We also extended our relativity studies to make an important contribution to relativistic geodesy with a new relativistic theory for falling-cube gravimeters that revealed previously unknown biases. Finally, we have modelled the effects of the ISS acceleration noise spectrum on the ISS as it orbits the Earth. Our objective is to determine the cause of the inconsistency we have seen in the variation of z-component of angular momentum and errors in coordinate transformations, systematic, or otherwise. Understanding this effect is key to the overall uncertainty imposed on the time transfer errors.
Bibliography Type:	Description: (Last Updated: 04/05/2023)
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